26F Stat TA Session W10

📢 Plan for Week 11

1. **No class** on Wednesday, Nov. 18
2. **Quiz 4** (Ch5-6) on Thursday, Nov. 19
3. We will have TA session after the quiz

1. Random Number Generator Revisited

- Today, we will be introduced to two more distributions: `Poisson` and `Exponential` (`Gamma`) distribution

`rpois()`

- Poisson distribution captures the distribution to answer the following question:

> ❓ Suppose that, on average, 2 buses arrive every hour. What is the probability that exactly one bus arrives within a 5-minute interval? (Number of Events)

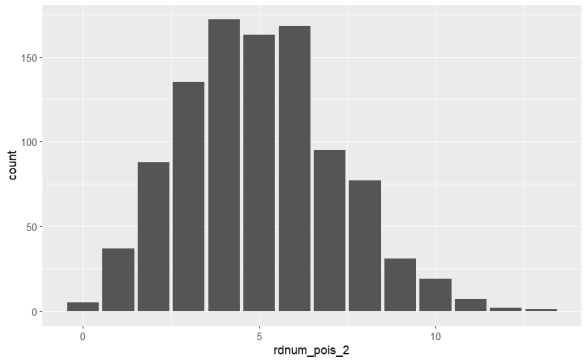


Figure: Random Samples with size 1000 drawn from Poisson($\lambda = 5$) (shown in bar plot)

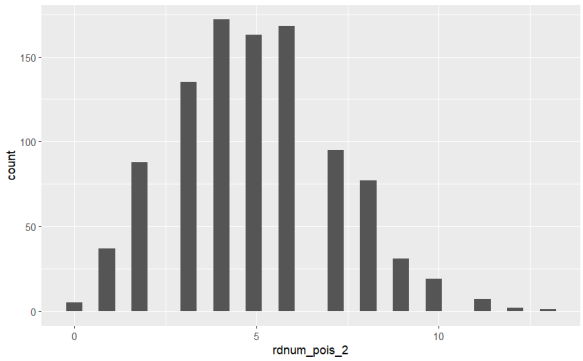


Figure: Random Samples with size 1000 drawn from Poisson($\lambda = 5$) (shown in histogram)

> ⚠️ Note that we use bar plot for discrete value counts (This is W3 material!)

```r
rdnum_pois <- rpois(20, lambda = 5)
```

```
> print(rdnum_pois)
 [1] 6 5 3 4 5 1 5 7 3 5 0 6 3 7 3 4 5 0 4 1
```

`rexp()`

- Similar to Poisson distribution, but in a reversed way. Exponential distribution answers:

> ❓ Suppose, similarly, there are on average 5 buses arrive each hour. Assuming bus arrivals follow a Poisson process, how long do you have to wait until the first bus arrives? (Waiting time)

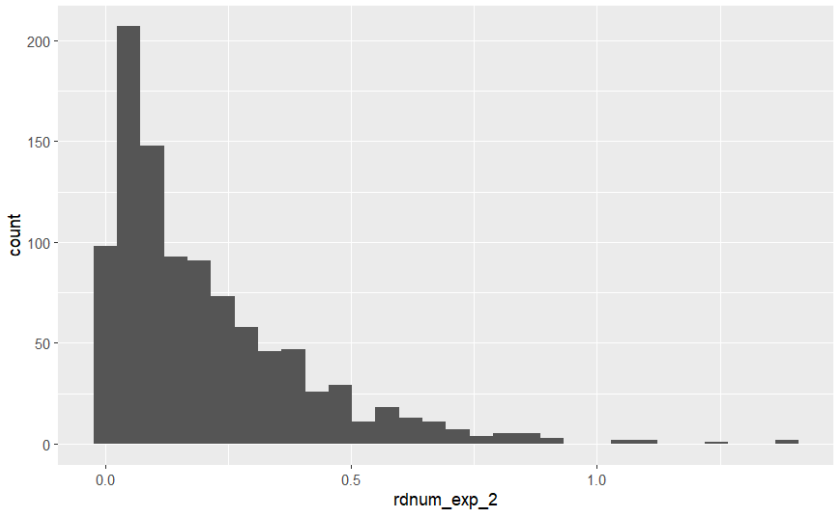


Figure: Random Samples with size 1000 drawn from Exp($\lambda = 5$)

```
rdnum_exp <- rexp(20, rate = 5)

> print(rdnum_exp)
 [1] 0.032277 0.647559 0.159807 0.063678 0.241689 0.081996 0.015779 0.016834
0.470081
[10] 0.145947 0.096140 0.112805 0.211684 0.161327 0.360304 0.365589 0.114410
0.402019
[19] 0.324005 0.369866
```

▼ (Optional) The Relation between `Poisson` and `Exponential` distribution

- You may have noticed that there is a close relation between the two distributions (number of buses coming vs waiting time for a bus)

📌 The c.d.f. of Poisson distribution is $F_X(x) = \frac{e^{-\lambda}\lambda^x}{x!}$, where $x = 0, 1, 2, \ldots$ and the mean is λ

- This captures the probability that within a time interval, x events occur (number of buses arrive)

💡 The key connection is "no arrival":
You have to wait a bus for t minutes = There is no bus arriving within t minutes

- Let T denote the time to the first bus arrival

➡️ $Pr(T > t) = Pr(X = 0|\mu = \lambda t) = e^{-\lambda t}$

- With simple differentiation of $Pr(T \leq t) = 1 - e^{-\lambda t}$, we attain $f_T(t) = \lambda e^{-\lambda t}$, which is the p.d.f. of exponential distribution!

2. Combine your Datasets

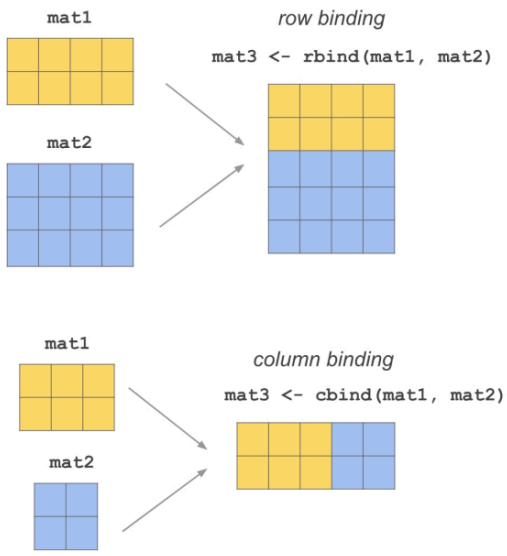


Figure: Combine dataset by rows or columns
(Gaston Sanchez, 2024)

Combine Dataset by Rows `rbind()`

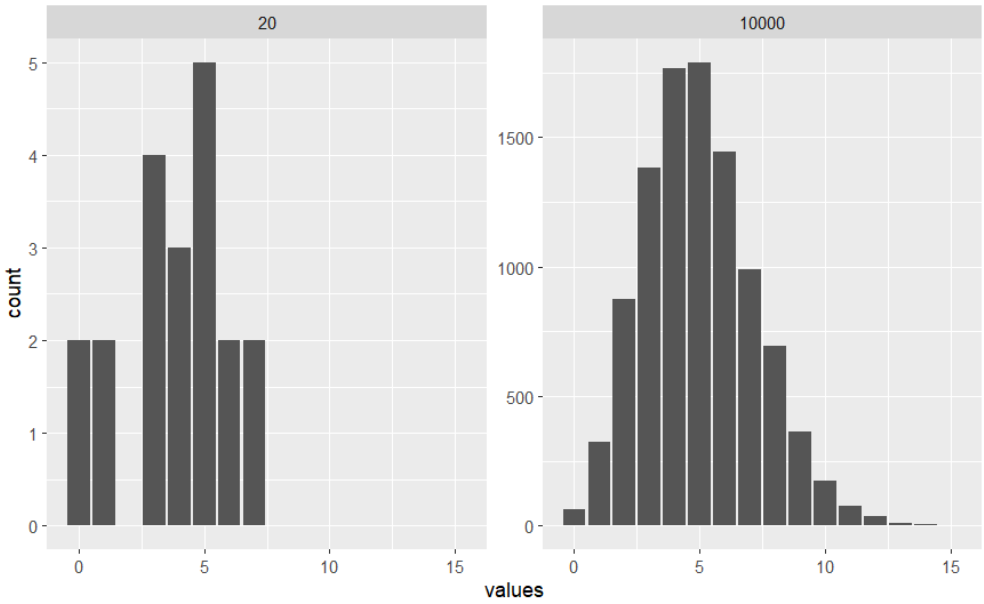


Figure: Histogram of two samples in different sample sizes

```r
# remember to set seed!
set.seed(2026)

# first, let's draw two different sets of random sample
rdnum_pois_20 <- rpois(n = 20, lambda = 5)
rdnum_pois_10000 <- rpois(n = 10000, lambda = 5)

# using the two sets of random sample, build two DataFrames respectively
df_20 <- data.frame(       # df_20 is 20x2
  size = 20,
  values = rdnum_pois_20
)
df_10000 <- data.frame(    # df_10000 is 10000x2
  size = 10000,
  values = rdnum_pois_10000
)

# combine the two DataFrames together, and draw the histogram
df <- rbind(df_20, df_10000)

ggplot(df, aes(x = values)) +
  geom_bar() +
  facet_wrap(~size, scales = "free_y")
  # facet_wrap is used when separating different groups for certain column
```

Combine Datasets by Columns `cbind()`

- This one is similar! One caveat is to check if the way of combination is valid
  - Use `View()` to check what the datasets look like before working on them! (or you can quickly check the environment on the right above section of `RStudio`)

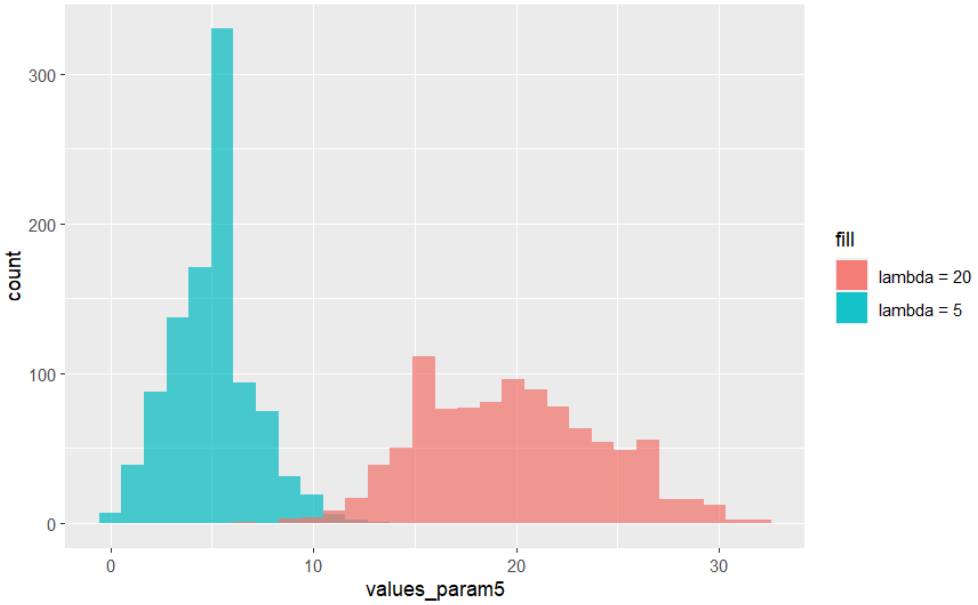


Figure: Histogram of two samples in different distribution parameters

```
rdnum_pois_param5 <- rpois(n = 1000, lambda = 5)
rdnum_pois_param20 <- rpois(n = 1000, lambda = 20)

df_param5 <- data.frame(              # df_param5 is 1000x1
  values_param5 = rdnum_pois_param5
)
df_param20 <- data.frame(             # df_param20 is also 1000x1
  values_param20 = rdnum_pois_param20
)

df <- cbind(df_param5, df_param20)
ggplot(df) +
  geom_histogram(mapping = aes(x = values_param5,
                               fill = "lambda = 5"),
                 alpha = 0.7
                ) +
  geom_histogram(mapping = aes(x = values_param20,
                               fill = "lambda = 20"),
```

```
                    alpha = 0.7
                  )
```

Exercises

W10 Exercises